

Part A

1. If a hot object and a cold object are placed next to each other, heat energy will flow from the _____ object to the _____ object.
2. Heat energy can transfer from one object to another by _____, _____, or _____.

Part B

3. What is conduction? Give an example of where it occurs.

4. Why are the handles of the BBQ grill made of wood?



5. Describe the basic process of how conduction works at the atomic level.

6. Give three examples of good conductors.

7. Give three examples of poor conductors.

8. The recipe for a cake states that it should be baked at 170°C for 45 minutes, but, feeling hungry, a student decides to set the oven to 250°C so that it can bake faster. The student removes the cake after 20 minutes when it has browned on top. What is the problem with this approach?

9. Why do surfers often wear wet suits?



10. People who are feeling cold will sometimes say “I need a nice warm blanket”, but blankets don’t produce heat. How, then do they keep us warm?

11. Why are solids better at conducting heat than liquids or gases?

Part C

12. What is convection?

13. Fan heaters and central heating units both heat up air that passes through them, but how does the warm air then get to the occupants of the room?

14. Even without being forced to, warm water rises (as does warm air), while cool water _____ (as does cool air).

15. Write the equation for density. _____
16. Fill in the table below. (working out space)

Temp (°C)	Mass (grams)	Volume	Density (g/cm ³)
0 (solid)	1	1.0929	
0 (liquid)	1	1.0001	
4	1	1.0000	1.0000
20	1	1.0018	0.9982
40	1	1.0078	
60	1	1.0171	
80	1	1.0290	
100 (liquid)	1	1.0434	0.9584

17. What happens to water's density as it gets warmer? _____
18. Why does warm water rise? _____
19. What is a convection current? Describe where you might find one. _____
20. Good insulation in a house is especially important above the ceiling. Why? _____
21. Your friend places two trays of muffins into an oven, one near the top and one near the middle, and is really surprised to find 30 minutes later that the muffins near the top have browned much more than the ones near the middle. How would you explain what happened to your friend? _____

Part D

22. Why can't the sun's heat energy reach Earth by conduction or convection? _____
23. How does the sun's heat reach the Earth? _____
24. What is an electromagnetic wave? _____
25. The electromagnetic spectrum consists of radio waves, _____, _____, _____, _____, and _____.
26. A 2400 W radiant heater does not appear very bright compared to a 10 W light globe. What is happening to all the electrical energy that that the radiant heater is using? _____
27. When invisible infrared light (or any form of electromagnetic radiation) is absorbed by an object, what happens? _____